

EXP 7: Write a Program to implement Sliding window protocol for Selective repeat

Aim:

To write and execute a program to implement the Sliding Window Protocol using Selective Repeat ARQ and demonstrate the retransmission of only the lost or erroneous frames.

Theory:

The Sliding Window Protocol is a flow control and error control mechanism used in computer networks. It allows the sender to transmit multiple frames before waiting for acknowledgements from the receiver. The number of frames that can be transmitted without receiving acknowledgements is determined by the window size.

Selective Repeat ARQ (Automatic Repeat request) is a sliding window protocol used for reliable data transmission. In this protocol, the receiver can accept and buffer frames that arrive out of order. Each frame is acknowledged individually.

When a frame is lost or corrupted during transmission, the receiver does not acknowledge that frame. The sender retransmits only the lost or erroneous frame, rather than retransmitting all the frames that follow it.

Algorithm:

Step 1: Initialization

1. Set total number of frames: TOTAL_FRAMES
2. Set window size: WINDOW_SIZE
3. Initialize sender window status array: sender_window[0 ... TOTAL_FRAMES-1] = False
4. Initialize receiver buffer: receiver_window[0 ... TOTAL_FRAMES-1] = None
5. Set base = 0 (index of the first frame in sender's window)

Step 2: Repeat Until All Frames Are Acknowledged

While base < TOTAL_FRAMES:

1. Set end = min(base + WINDOW_SIZE, TOTAL_FRAMES)
(marks the last index in current window)

Step 3: Send Frames in Current Window

For each i in range base to end - 1:

1. If sender_window[i] == False (i.e., not yet acknowledged):
 - Send frame i
 - Simulate random loss:
 - If frame not lost, store it in receiver_window[i]
 - Else, mark as lost (no change in receiver buffer)

Step 4: Process ACKs

For each i in range base to end - 1:

1. If receiver_window[i] is not None and sender_window[i] == False:
 - Assume ACK is received for frame i
 - Set sender_window[i] = True
2. Else:
 - ACK not yet received or frame lost

Step 5: Slide the Sender Window

While sender_window[base] == True and base < TOTAL_FRAMES:

- Increment base by 1

Step 6: Repeat

- Go back to Step 2 until base == TOTAL_FRAMES

Step 7: End

- Display: "All frames successfully transmitted and acknowledged."

Python Program:

```
import random
import time
# Constants
TOTAL_FRAMES = 10
WINDOW_SIZE = 4
LOSS_PROBABILITY = 0.3
DELAY = 0.8 # Time delay for demonstration

# Status arrays
sender_window = [False] * TOTAL_FRAMES # True means ACK received
receiver_window = [None] * TOTAL_FRAMES # None means not yet received
def selective_repeat_protocol():
    base = 0
    print("=== Selective Repeat Sliding Window Protocol ===\n")
    while base < TOTAL_FRAMES:
```

```

end = min(base + WINDOW_SIZE, TOTAL_FRAMES)
print(f'Sender Window: [{base} to {end - 1}]')
# Send all frames in the window
for i in range(base, end):
    if not sender_window[i]:
        print(f'Sending Frame {i}...', end=' ')
        if random.random() > LOSS_PROBABILITY:
            receiver_window[i] = f'Frame {i} received'
            print("Delivered.")
        else:
            print("Lost.")

# Process ACKs
for i in range(base, end):
    if receiver_window[i] is not None and not sender_window[i]:
        print(f'ACK received for Frame {i}')
        sender_window[i] = True
    elif receiver_window[i] is None:
        print(f'Waiting for ACK: Frame {i}')

# Slide the window
while base < TOTAL_FRAMES and sender_window[base]:
    base += 1
print("-" * 40)
time.sleep(DELAY)
print("\nAll frames successfully transmitted and acknowledged.")
# Run the simulation
selective_repeat_protocol()

```

Output: